

June 17, 2026

Dear PhD Admissions Committee,

I am writing to express my strong interest in pursuing a PhD in Quantum Computing and Optimisation for Large-Scale Energy System Planning under Uncertainty at the University of Southampton. As a data science engineer with a passion for applying AI to complex problems, I believe that this project aligns perfectly with my skills and interests.

My recent work on Medical<sub>MambaTS</sub> has granted me a unique perspective on the importance of flow-latency state space models in high-stakes applications. In the context of energy system planning under uncertainty, time state space models, we can develop scalable algorithms that account for uncertainty in renewable generation.

In addition to my experience with Medical<sub>MambaTS</sub>, I have also worked on  $PI_{DeepONet}$ , which has given me a deep understanding of Windkessel Physics-Informed Operator Networks that would allow us to develop more accurate and scalable energy system planning.

I am excited about the opportunity to join the University of Southampton as a PhD student and contribute my skills and experience to this innovative project. Thank you for considering my application.

Sincerely,

Ismaïl Aïssaoûi